

ITM SLS BARODA UNIVERSITY, VADODARA
B.TECH. SEMESTER-3 (BRANCH – ALL; SCHOOL – SOCSET)
DEPARTMENT OF APPLIED MATHEMATICS (S.O.S.)
COMPUTER ORIENTED NUMERICAL METHODS WITH PYTHON (S2320C1)

CET-1 SYLLABUS

Topic – Errors and Approximations

- ▶ Mathematical Preliminaries
- ▶ Accuracy of Numbers
 - Exact Numbers
 - Approximate Numbers
 - Significant Digits
 - Rounding-off Digits
 - Truncating Digits
- ▶ Errors
 - Definition
 - Types of Errors
 - Inherent Errors
 - Rounding-off Errors
 - Truncation Errors
 - Absolute, Relative, Percentage Errors

Topic – Solution of Algebraic and Transcendental Equations

- ▶ Mathematical Preliminaries
- ▶ Bracketing Methods
 - Bisection Method
 - Regula-Falsi Method
- ▶ Open-Ended Methods
 - Secant Method
 - Successive Approximation Method
 - Newton-Raphson Method & Deductions
 - Descartes' Rule of Signs
 - Budan's Theorem
 - Lin-Bairstow Method

Topic – Solution to System of Linear Equations

- ▶ Mathematical Preliminaries
- ▶ Iterative Methods
 - Jacobi Method
 - Seidel Method